Aiza Kabeer

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Education

PhD, Applied Mathematics | *University of Arizona, Tucson, AZ* *2023 –*
MS, Mathematics | *Drexel University, Philadelphia, PA* *2020 – 2022*
BS, Mathematics, Minor in Scientific Computation | *Simmons College, Boston, MA* *2012 – 2016*

Professional Experience

Department of Mathematics, University of Arizona *Tucson, AZ*
Graduate Outreach Scholar *Aug 2024 - Dec 2024*

- Supported the Center for Recruitment and Retention's events, such as Community Game Night, to foster local community engagement in mathematics
- Served on a selection committee to evaluate proposals for the Mathematics Educators Appreciation Day conference

School of Natural Resources and the Environment, University of Arizona *Tucson, AZ*
Summer Graduate Research Assistant *May 2024 - Aug 2024*

- Developed a novel analysis framework using unsupervised clustering algorithms to compare biomass products
- Performed an evaluation using geospatial biomass data from seven products to investigate variation by ecosystem type

Woodwell Climate Research Center *Falmouth, MA*
Research Assistant *Oct 2022 - Aug 2023*

- Applied a biogeochemical ecosystem model to understand carbon emissions from permafrost thaw
- Explored optimal model parameters by performing model-data calibration and sensitivity analysis
- Processed and interpreted model outputs using Python and Julia, while maintaining version control with Git

Optics @ Martinos, Martinos Center for Biomedical Imaging, Massachusetts General Hospital *Charlestown, MA*
Research Assistant *Aug 2018 - July 2020*

- Utilized numerical modeling of an inverse problem for breast cancer imaging projects in diffuse optical tomography
- Developed code to address complex research questions and support research studies
- Configured Arduino systems for elastography measurements and supported operations in human subject studies
- Tested and optimized imaging systems to refine hardware and algorithm performance, while managing IRB compliance

Draeger Medical Systems Inc *Andover, MA*
Programmer Analyst *Sept 2016 - May 2017*

- Programmed business applications for medical devices with SAP ABAP object oriented programming
- Analyzed, tested, and debugged configuration logic to ensure accurate functionality

Extracurricular Positions

- **Secretary, American Women in Mathematics** | University of Arizona (2026-2027)

Skills

Coding/Technical Skills Python, C, Matlab, Julia
Software Jupyter Notebooks, Git, ArcGIS, Linux, LaTeX, Docker, Arduino, MS Visio
Languages Urdu

Conference Papers and Publications

Kabeer, Aiza and McGuire, Luke (2026). “A numerical modeling investigation of rainfall intensity-duration thresholds for debris-flow initiation in a recently burned watershed in northern Arizona, USA”. In: *Earth Surface Processes and Landforms*. In review.

Kabeer, Aiza, Li, Yang, Scott, Eric, Wang, Jonathan A, Green, Julia K., Devine, Charles, and Moore, David JP (2024). “Using Clustering Analysis to Compare Satellite-based Aboveground Biomass Data Products”. In: *AGU Meeting Fall 24*. Poster Presentation.

Jafarov, Elchin E, Briones, Valeria, Sanderman, Jonathan, Genet, Helene, Vesselinov, Velimir monty V, Rogers, Brendan M, Natali, Susan, and **Kabeer, Aiza** (2023). “Semi-Automated Calibration of Terrestrial Ecosystem Model Parameters”. In: *AGU Meeting Fall 23*.

Muldoon, Ailis, **Kabeer, Aiza**, Cormier, Jayne, Saksena, Mansi A., Fang, Qianqian, Carp, Stefan A., and Deng, Bin (2022). “Method to improve the localization accuracy and contrast recovery of lesions in separately acquired X-ray and diffuse optical tomographic breast imaging”. In: *Biomed. Opt. Express* 13, pp. 5295–5310. DOI: 10.1364/BOE.470373.

Kabeer, Aiza, Saksena, Mansi A., Cormier, Jayne, Carp, Stefan A., and Deng, Bin (2020). “Compression-induced hemodynamic features of tumors in breast cancer patients undergoing neoadjuvant therapy: a longitudinal case study”. In: *Biophotonics Congress: Biomedical Optics 2020 (Translational, Microscopy, OCT, OTS, BRAIN), OSA Technical Digest*. Optica Publishing Group, paper JTh2A.7. DOI: 10.1364/TRANSLATIONAL.2020.JTh2A.7.

Kabeer, Aiza and Tsai, Jessica W. (2018). “Rectify biased interpretation of science history”. In: *Nature* 560, p. 29. DOI: 10.1038/d41586-018-05848-w.